EART26201 | Principles of Geochemistry | Practical 4

Applying phase diagrams to magmatic systems FEEDBACK

Task

This practical contains two questions about applying binary phase diagrams to understand magmatic systems. The first question should take about 40 minutes and the second question should take about 80 minutes.

I strongly encourage you to discuss your work with each other as you go.

Demonstrators will be available throughout the practical class, and I will provide a short round-up at the end of the session. Feedback will be posted on Blackboard in due course.

Please share any comments you have as we go along, and I will do my best to adjust.

I suggest typing your answers into this document as well as annotating the associated blank phase diagrams.

Goals

At the end of this practical you should be able to:

- Label the solidus, liquidus and phase fields on a binary phase diagram
- Interpret simple binary phase diagrams with no solid solution
- Interpret binary phase diagrams with a peritectic
- Use phase diagrams to explain equilibrium and fractional crystallisation and melting
- Use the lever rule to determine phase proportions on different phase diagrams

Other themes relevant to this week's content are reviewed in a Blackboard quiz.

You will revisit many of these themes in EART22200 (Igneous Minerals and Processes) and EART24200 (Metamorphic Minerals and Processes) later in the year.

Question 1: Interpreting a binary phase diagram with no solid solution (40 minutes)

- You should have a copy of this anorthite-diopside phase diagram on a separate sheet:

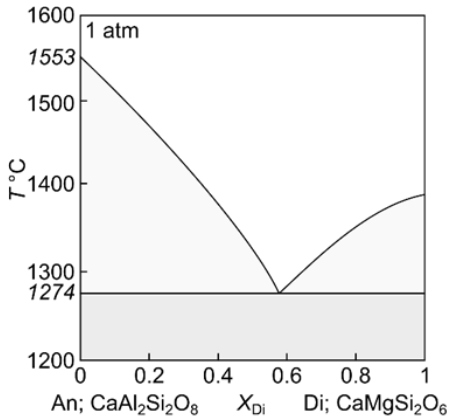


Figure 1: Anorthite-diopside phase diagram

a) Label the following:

i. The liquidus and solidus.

ii. The *L*, *L* + *An*, *L* + *Di* and *An* + *Di* phase fields.

iii. The eutectic point described by the reaction *L* = *An* + *Di*.

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b) Mark the composition X_{Di} = 0.8 on the phase diagram (i.e. 20% anorthite component and 80% diopside component).

c) Quantitatively describe the following scenarios:

i. Equilibrium crystallisation of a liquid with X_{Di} = 0.8.

1. The first crystals of Di form at ~1355 °C
2. The liquid composition then evolves along the liquidus as Di crystallises, becoming progressively lower in Di component
3. When the liquid composition reaches the eutectic at 1274 °C, the system contains liquid and Di proportions of 45% liquid and 55% Di. The liquid has the composition of the eutectic (XDi = 0.58)
4. Crystallisation continues at the eutectic, producing An and Di in proportions of 42% An and 58% Di until the liquid is exhausted
5. Once the liquid has been exhausted, the solid comprises An and Di in the proportions 20% An and 80% Di

ii. Fractional crystallisation of a liquid with X_{Di} = 0.8.

1. The first crystals of Di form at ~1355 °C and are removed from the system.
2. The liquid composition evolves along the liquidus as Di crystallises, becoming progressively lower in Di component; the composition of the system is the same as the composition of the liquid
3. When the liquid composition reaches the eutectic at 1274 °C, the system contains only liquid of the eutectic composition (XDi = 0.58).
4. Crystallisation continues at the eutectic, producing An and Di in proportions of 42% An and 58% Di until the liquid is exhausted.
5. The last solid to form contains An and Di in proportions 42% An and 58% Di

iii. Fractional melting of a solid with X_{Di} = 0.8.

1. The first liquid forms at the eutectic at 1274 °C, has a composition of XDi = 0.58 and is removed from the system
2. Melting continues at the eutectic while the solid residue becomes progressively lower in An component; the composition of the system is the same as the composition of the solid
3. Eutectic melting stops once the solid has a composition of Di
4. No melting occurs while the system heats from 1274 °C to ~1390 °C
5. Once at ~1390 °C, the remaining solid Di melts to form liquid of Di composition until all the solid has melted

d) How could you distinguish between rocks produced during fractional crystallisation of liquids with initial compositions of X_{Di} = 0.1 and X_{Di} = 0.8 in the field?

Fractional crystallisation of XDi = 0.1 would produce rocks of pure anorthite and then rocks with a mixture of anorthite and diopside. The rocks of pure anorthite would be more voluminous than the mixed rocks because the bulk composition is nearer anorthite than the eutectic.

Fractional crystallisation of XDi = 0.9 would produce rocks of pure diopside and then rocks with a mixture of anorthite and diopside. The rocks of pure diopside would be more voluminous than the mixed rocks because the bulk composition is nearer diopside than the eutectic. However, the eutectic is nearer XDi = 0.9 than XDi = 0.1, so there would be relatively fewer diopside-only rocks in the XDi = 0.9 case than anorthite-only rocks on the XDi = 0.1 case.

Question 2: Interpreting a binary phase diagram with a peritectic (80 minutes)

- You should have a copies of this ***zoomed in*** part of the forsterite-quartz phase diagram on a separate sheet (*remember that the right-hand-side is only 0.5 Qz and that En has a composition of* $X_{Qz} = 0.3$), as well as copies of the full forsterite-quartz phase diagram:

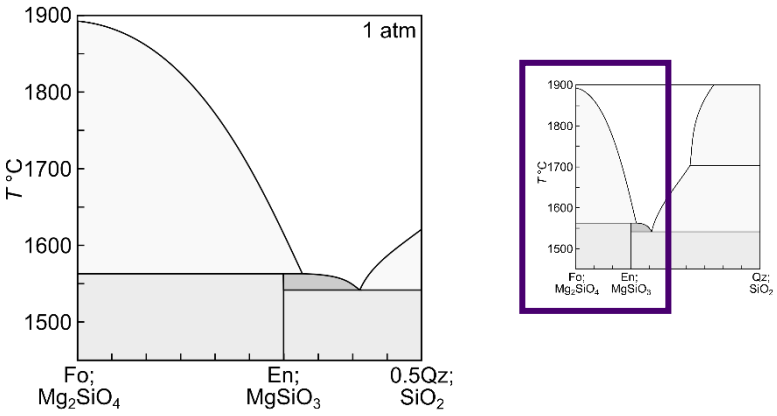


Figure 2: Forsterite-quartz phase diagram

a) Label the following:

i. The liquidus and solidus.

ii. The *L*, *L* + *Fo*, *L* + *En, L*+ *Qz*[*], *Fo* + *En* and *En* + *Qz* phase fields.

iii. The peritectic point described by the reaction *Fo* +*L* = *En*.

iv. The eutectic point described by the reaction *L* = *En* + *Qz*.

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b) Mark the composition X_{Qz} = 0.32, X_{Fo}= 0.68 on the ***zoomed in*** phase diagram (i.e. 32% quartz component and 68% forsterite component; *remember that the right-hand-side is only 0.5 Qz and that En has a composition of* $X_{Qz} = 0.3$). We are using the zoomed in diagram to see what happens around the peritectic with clarity.

[*] Strictly speaking, Cristobalite (Cr) is the stable SiO_2 polymorph under these temperature conditions,

c) Quantitatively describe the following scenarios:

i. Equilibrium crystallisation of a liquid with X_{Qz} = 0.32.

1. The first crystals of Fo form at ~1580 °C
2. The liquid composition then evolves along the liquidus as Fo crystallises, becoming richer in Qz component
3. When the liquid composition reaches the peritectic at ~1565 °C, the peritectic reaction occurs until the Fo has been consumed and only liquid and En remain.
4. The liquid composition then evolves along the liquidus as En crystallises, becoming richer in Qz component
5. When the liquid composition reaches the eutectic at ~1540 °C, the system contains liquid and En proportions of 20% liquid and 80% Di. The liquid has the eutectic composition of XQz = 0.42, XFo = 0.58
6. Crystallisation continues at the eutectic, producing En and Qz in proportions of 83% En and 17% Qz until the liquid is exhausted
7. Once the liquid has been exhausted, the solid comprises En and Qz in the proportions 97% En and 3% Qz; this is the same composition as the starting liquid and the system as a whole (XQz = 0.32, XFo = 0.68)

ii. Fractional crystallisation of a liquid with X_{Qz} = 0.32.

1. The first crystals of Fo form at ~1580 °C and are removed from the system
2. The liquid composition then evolves along the liquidus as Fo crystallises, becoming richer in Qz component
3. When the liquid composition reaches the peritectic at ~1565 °C, the peritectic reaction does not occur because there is no Fo (all Fo has been removed from the system); the composition of the system is the composition of the liquid
4. The liquid composition then evolves along the liquidus as En crystallises, becoming richer in Qz component. En crystals are removed from the system
5. When the liquid composition reaches the eutectic at ~1540 °C, the system contains only liquid of the eutectic composition (XQz = 0.42, XFo = 0.58)
6. Crystallisation continues at the eutectic, producing En and Qz in proportions of 83% En and 17% Qz until the liquid is exhausted.
7. The last solid to form contains En and Qz in proportions 83% En and 17% Qz

iii. Fractional melting of a solid with X_{Qz} = 0.32.

1. The first liquid forms at the eutectic at 1540 °C, has a composition of XQz = 0.32, XFo = 0.68 and is removed from the system
2. Melting continues at the eutectic while the remaining solid becomes progressively lower in Qz component; the composition of the system is the same as the composition of the solid
3. Eutectic melting stops once the solid has a composition of En
4. No melting occurs while the system heats from ~1540 °C to ~1565 °C
5. Once at ~1565 °C, the solid En melts incongruently to produce L and Fo
6. Melting at the peritectic stops once the solid En has been exhausted and only solid Fo remains; all liquid has been removed from the system
7. No melting occurs while the system heats from ~1565 °C to ~1890 °C
8. Once at ~1890 °C, solid Fo melts to form liquid of Fo composition until all the solid has melted

d) After discussion with your neighbours, briefly describe how each of the melting or crystallisation scenarios would differ if the system had a composition of X_{Qz} = 0.28, X_{Fo}= 0.72 instead of X_{Qz} = 0.32, X_{Fo}= 0.68.

Equilibrium crystallisation:

When the bulk composition lies between Fo and En instead of between En and Qz, L will be exhausted before Fo during the peritectic reaction Fo + L = En, meaning that the last liquid will crystallise at the peritectic and the solid residue will contain Fo and En.

Fractional crystallisation:

Because the peritectic reaction does not occur during fractional crystallisation, there will be no substantial change in the crystallisation behaviour. The only difference will be that slightly more Fo will crystallise from the liquid before the system composition passes through the peritectic point and En crystallises from the liquid instead of Fo.

Fractional melting:

The first liquid will form at the lowest temperature point of the phase diagram where Fo and En are in equilibrium with liquid. When the bulk composition lies between Fo and En instead of between En and Qz, the first liquid therefore forms at the peritectic rather than the eutectic.